SARS-CoV-2 Antibody Dynamics in Immunocompromised and Healthy Individuals

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Background

Understanding the longitudinal dynamics of SARS-CoV-2 antibody responses is critical for accurately interpreting seroprevalence estimates, assessing the durability of immunity, and informing vaccination strategies. Antibody trajectories are heterogeneous and shaped by prior infection, vaccination history, host characteristics, and clinical disease status. Robust characterization of these complex temporal patterns requires large-scale, harmonized serological data collected across diverse populations and settings.

Serological Sciences Network (SeroNet)

The SeroNet [1], launched by the National Cancer Institute in 2020 in response to the COVID-19 pandemic, is a nationwide consortium dedicated to advancing the scientific understanding of SARS-CoV-2 immune responses. The initiative brings together research institutions, capacity-building centers, and coordinating laboratories to develop, rigorously validate, standardize, and harmonize serological assays across platforms and sites. Through the establishment of reference standards and collaborative assay benchmarking, it enhances cross-study comparability and data integration. This coordinated infrastructure supports large-scale investigations of antibody kinetics, immune durability, and correlates of protection, providing a foundation for evidence-based public health and clinical decision-making.

Leidos Project (Prime: NIH)

"Multiplexed in-solution serological test for SARS-CoV-2, human coronaviruses and other respiratory pathogens" ($14.9M; October 2021 – September 2025) aimed to develop, validate, and implement a large-scale clinical serological assay within ASU Biodesign Clinical Testing Laboratory (ABCTL). This effort established the ASU Capacity Building Center for SARS-CoV-2 serological testing and partnered with clinical collaborators including Columbia University, Dignity Health, Valleywise Health, and Abrazo Community Health Network.

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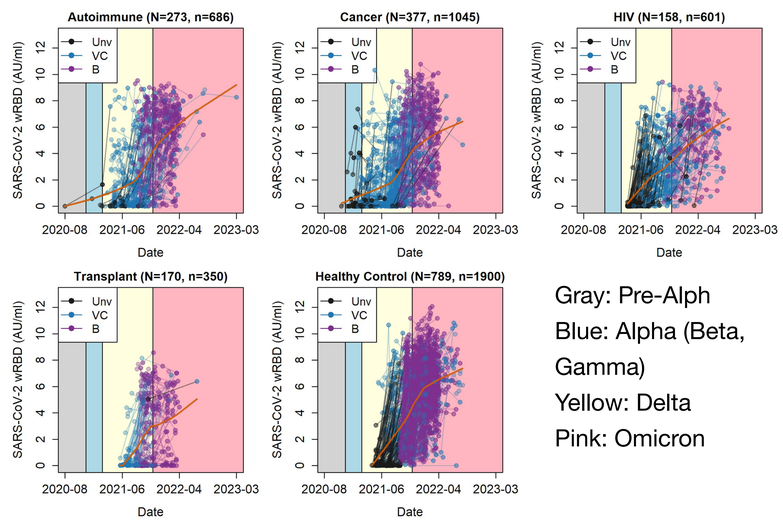
SARS-CoV-2 Data

A total of 1,767 participants contributed 4,582 longitudinal samples, collated across Arizona State University, University of Minnesota, Feinstein Institutes for Medical Research, and Icahn School of Medicine at Mount Sinai. Demographic information and vaccination histories were collected for each participant. Antibody responses against SARS-CoV-2 antigens and infection status were measured using multiplexed in-solution protein array (MISPA) [2]. The study population comprised five cohorts: individuals with autoimmune disease, cancer, HIV, solid organ transplant recipients, and healthy controls.

Overall Trend



Gray: Pre-Alph
Blue: Alpha (Beta, Gamma)
Yellow: Delta
Pink: Omicron

References

[1] Karger, A.B., Brien, J.D., Christen, J.M., Dhakal, S., Kemp, T.J., Klein, S.L., Pinto, L.A., Premkumar, L., Roback, J.D., Binder, R.A. and Boehme, K.W., 2022. The Serological Sciences Network (SeroNet) for COVID-19: depth and breadth of serology assays and plans for assay harmonization. MSphere, 7(4), pp.e00193-22.
[2] Ridha, I., Xu, C., Zhang, Y., Chung, Y., Park, J.G., LaBaer, J. and Murugan, V., 2024. Multiplexed methylated DNA immunoprecipitation sequencing (Mx-MeDIP-Seq) to study DNA methylation using low amounts of DNA. DNA, 4(4), pp.397-416.

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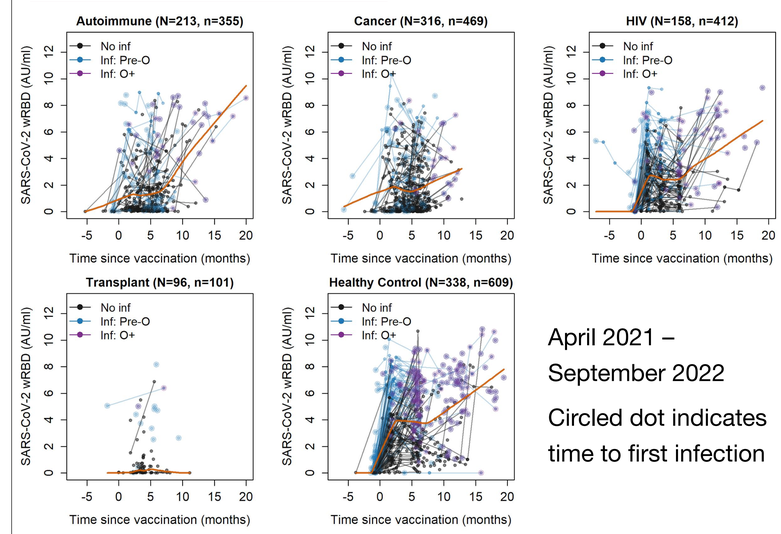
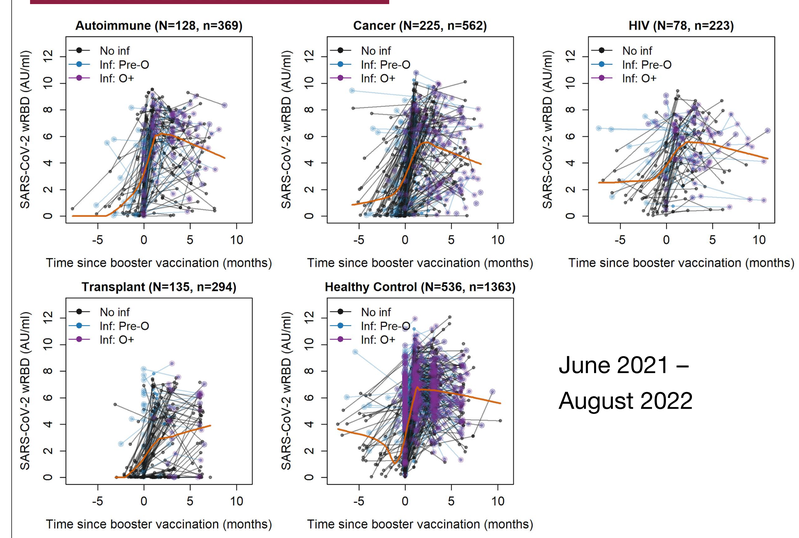
Trend after Vaccine Completion



April 2021 – September 2022

Circled dot indicates time to first infection

Trend after Booster Vaccination



June 2021 – August 2022

Statistical Modeling

- Nonlinear mixed-effect models were fitted with interaction terms between group × cubic spline time ($p<0.05$) and between vaccination × infection status ($p<0.05$).
- Peak time and slope decay varied across the five cohorts ($p<0.05$).

Future Research: Statistical Methodology Development

- Is the booster vaccine effective against SARS-CoV-2 Infection?
- Are these findings robust against diagnostic errors, measurement errors, and unmeasured confounders?